

Smart Irrigation Agent

Problem Formulation and Environment Classification

1. Problem Formulation

Element	Description
Initial State	The field condition is unspecified to the agent until its sensors report the starting readings for soil moisture, temperature, and the current weather outlook. No assumption is made in advance about whether the field needs water.
Actions	Turn the water pump ON; turn the water pump OFF; open or close the valve for a given field section; hold and continue monitoring (take no action this cycle); alert the farmer of the current pump status.
Transition Model	Each time the sensors report new readings, or the weather forecast is refreshed, the agent's internal picture of the soil condition, its pump/valve status, and its confidence in the current decision are all updated accordingly.
Path Cost	Cost is measured in terms of water consumed, pump running time/energy used, and the number of monitoring cycles carried out. The aim is to keep the soil within its ideal moisture range while using the least amount of water and power possible.
Goal Test	Success is reached when the soil moisture for the field stays within the optimal range for the crop, water is supplied only when genuinely required, and the pump/valve status correctly matches the field's real need at all times.
State Space	All possible combinations of sensor readings (moisture level, temperature, forecast outcome), pump status (ON/OFF), valve status, and the timing of the last irrigation cycle.
Search Space	All possible sequences of sensing conditions, evaluating whether irrigation is needed, acting on the pump/valve, waiting through an interval, and re-checking the field — repeated continuously as the season progresses.
Agents	Single agent. The irrigation controller acts alone on behalf of the farmer; the task involves no other agent that competes or cooperates with it.

2. Environmental Properties

Property	Classification	Why
Observability	Partially Observable	The agent perceives only soil moisture, temperature, and the weather forecast. It cannot directly observe deeper conditions such as root health, underground water table, or actual plant stress, so its picture of the field is incomplete.

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Determinism	Stochastic / Non-deterministic	Turning the pump ON does not guarantee a predictable rise in soil moisture — unexpected rainfall, a sudden temperature swing, or a sensor/equipment fault can all change the actual outcome.
Episodes	Sequential	Each irrigation decision affects the state the agent will face afterward; watering heavily now changes what the soil moisture reading will be in the next cycle, so past actions shape future ones.
Change	Dynamic	The field keeps changing independently of the agent's decisions — the sun evaporates moisture, the temperature rises through the day, and weather shifts, all while the agent is still deciding.
Values	Discrete & Continuous	The pump and valve states (ON/OFF, open/closed) are discrete, while moisture percentage, temperature, and water flow rate are continuous quantities that can take any value within a range.
Agents	Single Agent	Only the irrigation controller senses and acts in this environment; no second agent shares or contests control of the field.

3. Choosing the State Type

Option	Suitability	Explanation
Multi-state	Most suitable	The agent passes through many distinct states as it operates: raw sensor readings, the evaluated soil condition, the pump/valve decision, the action taken, and the updated field state after that action. A single-state or fixed-schedule model cannot capture this ongoing cycle of sensing, deciding, and re-checking.

Final Answer: The Smart Irrigation Agent is best classified as a Multi-state agent because it continuously moves between changing states — sensing the field, evaluating soil and weather conditions, deciding on a pump/valve action, and updating its knowledge once the action is taken. Its environment is also partially observable and stochastic, since the agent cannot directly observe every condition affecting the crop and cannot guarantee an exact outcome from any single watering decision.